COURSE END PROJECT

Automatic Fan Control Using IR Sensor

IOT LABORATORY (A8603)

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

SUBMITTED BY

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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This is to certify that the Course End Project titled **"Automatic Fan Control Using IR Sensor"** is carried out by **M.Durgaprasad** and **T.Akash** with Roll Numbers **24881A05G6 and 24881A05J8**towards **A8603 - IOT Laboratory** course in partial fulfilment of the requirements for the award of degree of **Bachelor of Technology** in **Computer Science and Engineering** during the Academic year 2025-26.

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Check List

S. No.	Content	Status (✓ / ×)
1	Abstract	✓
2	Objectives	✓
3	Problem Statement	✓
4	Algorithm	✓
5	Flow Chart	✓
6	Source Code	✓
7	Input	✓
8	Output	✓
9	Conclusion	✓
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11	References	✓

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1.ABSTRACT

This project is about making a fan that turns on and off by itself. Usually, people forget to turn off fans when they leave a room, which wastes electricity. This system solves that problem by using an IR (Infrared) sensor to detect when someone is there.

The heart of the project is an Arduino Mega. We connected an IR sensor to the Arduino to act as the "eyes" of the system. When the sensor sees a person or an object, it sends a signal to the Arduino. The Arduino then tells a Relay to switch the fan on. When the person moves away and the sensor stops seeing them, the Arduino automatically switches the fan off.

We used a Relay because the Arduino cannot power a fan directly without getting damaged. The relay acts like a safe middle-man switch.

This project is great because it is cheap to build and easy to install in homes or offices. It makes life more convenient because you don't have to touch a switch, and it helps save money on electricity bills by making sure the fan only runs when it’s actually needed. In the future, we can add more sensors or even control it with a smartphone.

Main Parts Used:

- **Arduino Mega:** The brain that controls everything.
- **IR Sensor:** To detect movement or presence.
- **Relay Module:** To safely turn the fan motor on and off.
- **Fan:** The device being controlled.

2. PROBLEM STATEMENT

The traditional way of operating fans in homes and offices relies entirely on manual switches, which leads to several practical issues. The biggest problem is human forgetfulness; people often leave a room without turning off the fan, causing it to run for hours in an empty space. This results in a significant waste of electrical energy and leads to unnecessarily high electricity bills. In a large building or a college campus, this cumulative waste of power is a major concern that needs a more efficient solution than just relying on people to remember to flip a switch.

Another issue with manual control is the constant physical effort required, which can be inconvenient in busy environments. In some cases, reaching for a switchboard may not be easy for everyone, and constant manual switching can lead to wear and tear on the electrical components over time. There is also a small but relevant concern regarding hygiene and ease of use in public or shared spaces where multiple people touch the same switches throughout the day.

To solve these challenges, there is a clear need for an automated system that can handle fan operation independently. By using an IR sensor and an Arduino Mega, we can create a responsive environment where the fan detects the presence of a person and switches on automatically. This eliminates the need for human intervention, ensures that electricity is only consumed when the room is actually occupied, and provides a modern, hands-free experience that is both cost-effective and easy to maintain.

Furthermore, the implementation of this system provides a layer of safety and durability that manual switches lack. Mechanical switches are prone to sparking and internal damage over time due to constant physical use, whereas a microcontroller-based system uses electronic signals to manage the load. By integrating a relay with the Arduino Mega, the high-voltage circuit is kept separate from the user, reducing the risk of electrical accidents. This automation not only makes the environment smarter but also extends the lifespan of the electrical setup by ensuring the fan is never overworked when the room is vacant.

Beyond the energy savings, this automated setup also represents a step toward modernizing infrastructure without requiring expensive smart-home installations. Traditional electrical systems are static and "blind" to their surroundings, but adding an IR sensor and a microcontroller creates a responsive environment that adapts to human movement. This makes the system particularly useful in public areas like hallways, shared restrooms, or library cabins where people are constantly coming and going. By removing the dependency on the last person leaving a room to turn off the power, the system creates a more sustainable and reliable method for managing common electrical appliances.

3. OBJECTIVES

The main objective of this project is to build an automated fan control system that works without any human help. By using an Arduino Mega and an IR sensor, the system aims to detect when a person is nearby and immediately turn the fan on. This removes the need for someone to manually walk over to a switchboard every time they enter or leave a space, making the environment more responsive and modern.

Another important goal is to focus on energy conservation. In many cases, fans are left running in empty rooms because of forgetfulness, which leads to a lot of wasted electricity. This project aims to solve that by ensuring the fan only draws power when the IR sensor confirms that the room is occupied. As soon as the person leaves and the signal is lost, the Arduino is programmed to cut the power, helping to lower electricity costs and save energy.

The main objectives of the project are:

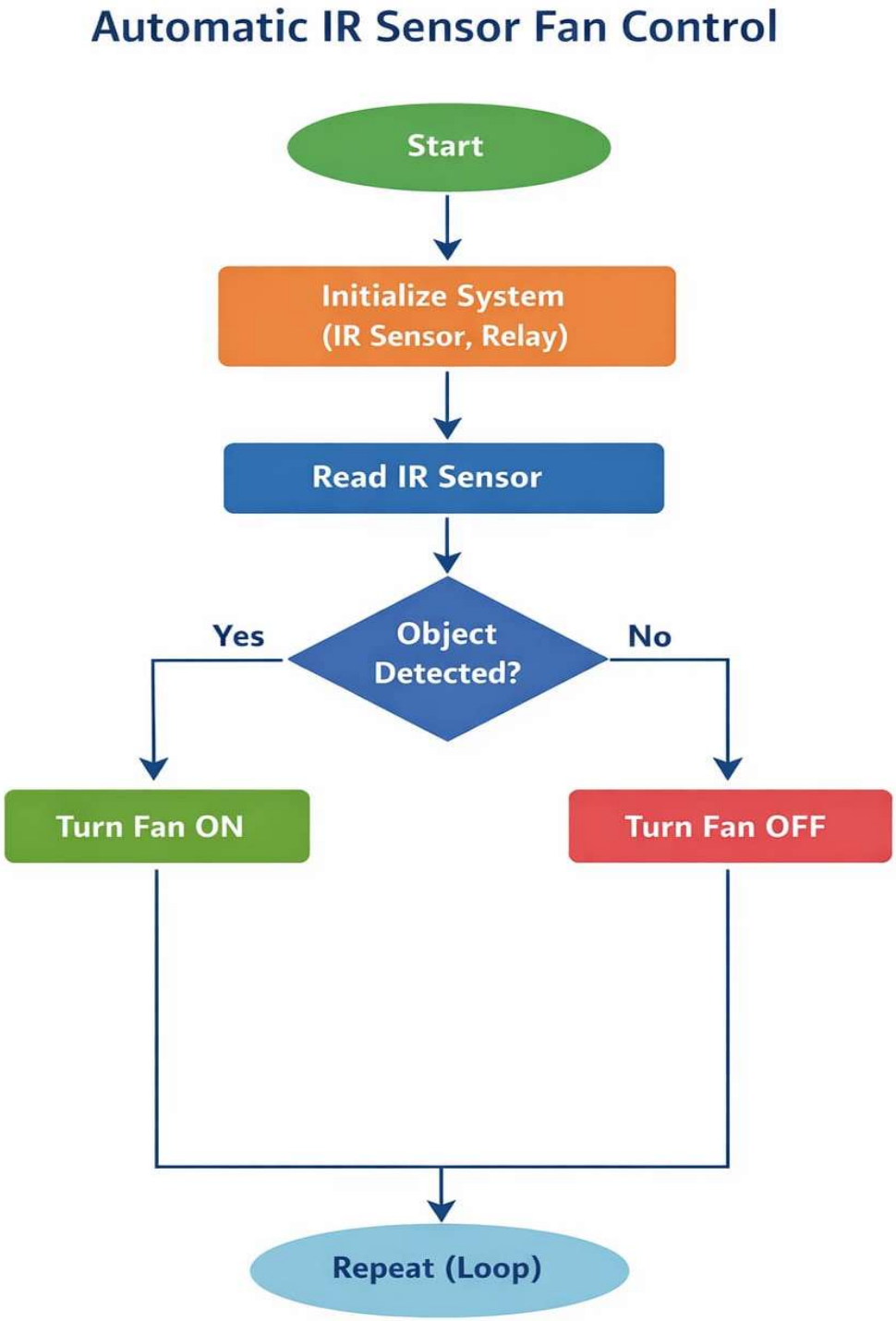
- To design an automatic cooling system for hands-free fan operation.
- To detect the presence of a person or an object using an IR proximity sensor.
- To develop a control mechanism that automatically switches the fan on and off.
- To reduce the need for manual operation of electrical switches.
- To minimize the wastage of electricity caused by fans left running in empty rooms.
- To improve energy efficiency in residential, office, and college environments.
- To ensure safe operation of high-power loads using a relay and an Arduino Mega.
- To demonstrate the practical application of sensor-based automation in everyday life.
- To create a low-cost and easy-to-implement alternative to expensive smart home systems.

4. Algorithm

1. **Start** the system by providing power to the Arduino Mega.
2. **Initialize** all the components, including the IR proximity sensor, the 5V relay module (acting as the fan switch), and the Serial Monitor for debugging.
3. **Set the initial condition** where the relay is deactivated and the fan is in the **OFF** position.
4. **Continuously monitor** the digital output signal from the IR sensor.
5. **Detect** whether a person or object is present in front of the sensor:
6. **Logic High (1)** → No object detected (Infrared light is not reflected).
7. **Logic Low (0)** → Object detected (Infrared light is reflected back to the sensor).
8. **Evaluate the sensor state** based on the input signal.
9. **If a person is detected** (Sensor signal is Low):
10. Send a trigger signal to the Relay module.
11. Close the high-voltage circuit to turn the **Fan ON**.
12. **If no person is detected** (Sensor signal is High):
13. Send a stop signal to the Relay module.
14. Open the high-voltage circuit to turn the **Fan OFF**.
15. **Maintain the state** as long as the sensor reading remains constant.
16. **Implement a short delay** (e.g., 200ms) to stabilize the system and prevent the relay from "chattering" or switching too rapidly due to signal noise.
17. **Repeat the process** indefinitely to ensure real-time automated operation.
18. **Stop** the system only when the main power supply is disconnected.

5. PROPOSED SYSTEM

5.1 FLOW CHART



5.2 WORKING MODEL

1. Power Supply Initialization
The system starts with a power supply (USB or battery). All components such as the IR sensor, Arduino Uno, and fan driver circuit are initialized and ready to operate.
2. Human/Object Detection
An IR sensor is placed in the room facing the detection area. It continuously emits infrared rays and checks for reflections.
=> When no person is present → sensor output is HIGH
=> When a person approaches → sensor output becomes LOW
3. Signal Processing
The IR sensor sends detection data to the Arduino. The Arduino processes this signal and checks whether someone is present within the detection range.
4. Fan Control Mechanism
Based on the processed data:
=> If no person is detected → Fan remains OFF
=> If a person is detected → Fan turns ON automatically
The Arduino activates a transistor or relay to supply power to the fan.
5. Automatic Turn OFF System
After the person leaves the detection area:
The sensor no longer detects movement
The Arduino waits for a short delay (optional)
Then the Arduino turns OFF the fan automatically
6. Alert / Indication System (Optional)
An LED can be used to indicate:
Fan ON condition
Fan OFF condition
System operating status
7. Continuous Monitoring
The system continuously monitors the room. For every person entering or leaving, the process repeats automatically.
8. IoT Extension (Optional)
The system can be upgraded using Wi-Fi modules like the ESP8266 to:
Monitor fan status remotely
Control fan using mobile app
Track usage time
9. Real-Time Operation
The entire system works in real time:
Quick response to human presence
Automatic switching of fan
No human intervention required

6.SOURCE CODE

```
const int irSensorPin = 2;
const int motorPin = 13;

void setup() {
  pinMode(irSensorPin, INPUT);
  pinMode(motorPin, OUTPUT);
}

void loop() {
  if (digitalRead(irSensorPin) == LOW) {   // Object detected
    digitalWrite(motorPin, HIGH);          // Motor ON
  } else {
    digitalWrite(motorPin, LOW);           // Motor OFF
  }
}
```

7.INPUT

The input to the automatic fan control system is provided through an **IR sensor**. This sensor plays a very important role as it detects the presence of a person in the monitoring area. It works by transmitting infrared rays and detecting the reflected rays when they strike an object such as a human body. When the reflected rays are received, the sensor changes its output signal.

The IR sensor continuously monitors the surroundings and sends its output to the microcontroller (Arduino Uno). The Arduino reads this signal and processes it to determine whether a person is present or not. The sensor generally provides a digital output, which makes it easy for the microcontroller to understand the condition.

• When no person is present → IR rays are not reflected back → sensor output remains HIGH → input condition = no object detected

• When a person comes near the sensor → IR rays are reflected back → sensor output becomes LOW → input condition = object detected

The Arduino continuously reads this input signal in real time using one of its digital input pins. Based on this information, it decides the operation of the fan. If the sensor detects a person, the Arduino activates the output pin to turn ON the fan. If no person is detected, the Arduino keeps the fan in OFF condition.

The detection range of the IR sensor can be adjusted using the onboard potentiometer available on the sensor module. This allows the system to be tuned according to room size and required sensing distance. The continuous monitoring ensures that the system responds quickly to any movement and operates automatically without human intervention.

8.OUTPUT

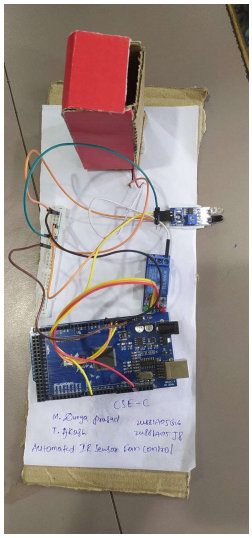
The output of the automatic fan control system is mainly controlled through the **fan motor**, along with optional indicators like LED or buzzer.

Based on the processed input (IR sensor signal), the system produces the following outputs:

• When the sensor detects a person (object detected):
o The fan motor turns ON automatically
o The LED turns ON (indicating fan operation)
o The buzzer may produce a short sound (alert indication – optional)
o This indicates that the system is actively cooling the area

• When the sensor does not detect a person:
o The fan motor remains OFF
o The LED remains OFF
o The buzzer remains OFF
o This indicates that no one is present in the detection area

After the person leaves the sensing range, the fan automatically turns OFF. A small delay can also be added to avoid frequent ON/OFF switching. The output keeps changing dynamically based on real-time sensor input, ensuring smooth and automatic operation without any manual effort.



9.CONCLUSION

The automatic fan control system is a simple yet effective solution that improves the way electrical appliances are managed. In many places, fans are still operated manually, which can lead to unnecessary power consumption, inconvenience, and wastage of energy when no one is present. This project provides a smart alternative by automating the entire process.

By using an IR sensor along with an Arduino microcontroller, the system can easily detect the presence of a person and operate the fan accordingly. When a person enters the detection area, the fan turns ON automatically, and when the person leaves, it turns OFF. This ensures efficient usage of electricity without requiring any manual switching.

One of the key advantages of this system is that it reduces manual effort and saves electrical energy. It also improves efficiency by ensuring that the fan operates only when needed. The system is reliable, fast, and easy to use, making it suitable for real-life applications.

Another important aspect of this project is that it demonstrates the practical use of automation and basic smart control concepts. Even though it is a simple model, it clearly shows how technology can be used to solve everyday problems in an efficient way.

The system is designed to be low-cost, easy to install, and user-friendly. It can be used in homes, classrooms, offices, and public spaces where automatic control of fans is beneficial. Overall, this project is a great example of how smart systems can make daily operations easier and more energy efficient.

In conclusion, the automatic fan control system highlights the importance of automation in modern life. It not only improves convenience but also contributes to energy saving and smarter management of electrical appliances.

10.FUTURE SCOPE

The automatic fan control system has a wide scope for further development and improvement. With the help of advanced technologies, this simple system can be transformed into a more intelligent and energy-efficient solution.

One major improvement is the addition of temperature sensors, which can automatically adjust the fan speed based on room temperature. This will ensure better comfort and efficient energy usage by running the fan only when required.

The system can also be enhanced by integrating a mobile application that allows users to monitor and control the fan remotely. This is especially useful in smart homes, offices, and classrooms where centralized control is needed.

Another possible upgrade is the use of motion detection using advanced sensors such as PIR sensors. This will improve detection accuracy and allow the system to distinguish between human presence and other objects.

The system can also be powered using solar energy, making it more energy-efficient and suitable for outdoor installations or areas with limited power supply.

In addition, data logging and analytics can be added to track fan usage patterns. This information can help in understanding energy consumption and improving efficiency.

The integration of IoT can further improve the system by enabling real-time monitoring, remote access, and cloud-based control. Users can check the fan status and control it from anywhere using the internet.

Overall, the future scope of this project is very broad. With continuous improvements, this system can be used in smart homes, smart classrooms, and smart buildings to provide better automation, energy saving, and efficient appliance management.

11.REFERENCES

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- • IR Sensor Module Guide
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- • IR Obstacle Sensor Datasheet
  (Used to understand detection range and signal output characteristics)
- • Electronics Tutorials – https://www.electronicshub.org
  (Basic concepts of sensors, automation, and control circuits)
- • Embedded Systems and Arduino Textbooks
  (For theoretical knowledge of microcontrollers and automation systems)
- • YouTube Tutorials (Arduino Automatic Fan Projects)
  (For practical implementation and circuit connections)
- • Research Articles on Smart Home Automation Systems
  (For understanding real-world applications of automatic fan control and energy saving)